

Summer Term 2026 – Julián Méndez & Ricardo Langner

Interactive Information Visualization

Exercise: **Assignment 3**

Introduction Assignment 3

Dataset

For this assignment, you will use a graph of soccer players from the 2017/18 Champions League season. A JSON file of this dataset is included in the programming template.

The dataset consists of:

- 95 soccer players (nodes), for each player up to 40 data attributes
- ~1050 relations (edges) between players, an edge indicates if and how often the players have played for the same club during their career (edge weight, val)

Note: This was used in Responsive Matrix Cells (<https://imld.de/rmc>)

Introduction Assignment 3

Programming Template

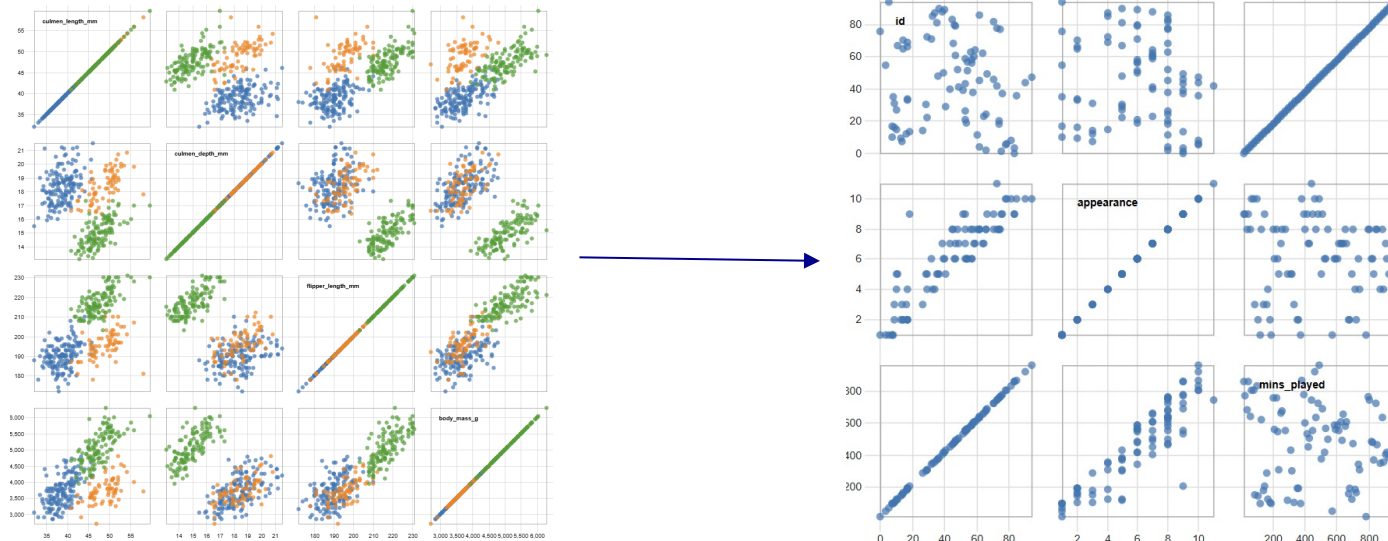
- We will use JavaScript and D3.js (<https://d3js.org>).
- We only scratch the surface, but D3 has a few unique features and thus comes with *a bit* of a learning curve. Check some tutorials online, if you're new!
- We prepared a programming template that you will fork to work on your solutions. It loads all the necessary dependencies for you already (D3.js, Bootstrap, the soccer dataset, etc.).

Template: <https://github.com/menocsk27/infovis-a3>

Multivariate Visualization in Multiple Coordinated Views

Adjust a Scatterplot Matrix

- Based on this example: <https://observablehq.com/@d3/brushable-scatterplot-matrix>
- Load the soccer dataset (you can load a subset of attributes)
- Invert the diagonal (i.e., [/] instead of [\]).



Multivariate Visualization in Multiple Coordinated Views

Linked Brushing

- Note how both the PCP and the SPLOM have brushing functionalities.
- Synchronize them! A selection in the PCP should e.g., change the color of the selected data points in the SPLOM.
- **Hint:** use *CustomEvents* :)

When submitting your solutions:

- Same as the other assignments, except simpler:
 - Upload a .txt with the names of your team members and the link to your (public) fork.
 - If you don't want your solution public for some reason, upload a .zip file with the .txt and your sources.
- You can again use AI, if you wish.
 - However, keep you should be able to read / execute code (for the exam). So, don't rely on AI excessively.
 - If you use AI, tell us also: which model did you use? How?
- Have fun!

For Assignment 4:

...the soccer players dataset is a *network*,
we will consider edges/links next time :)

Exercise Interactive Information Visualization

Thank you!

Any questions?

Contact: julian.mendez2@tu-dresden.de
ricardo.langner@tu-dresden.de